

CSE 311:

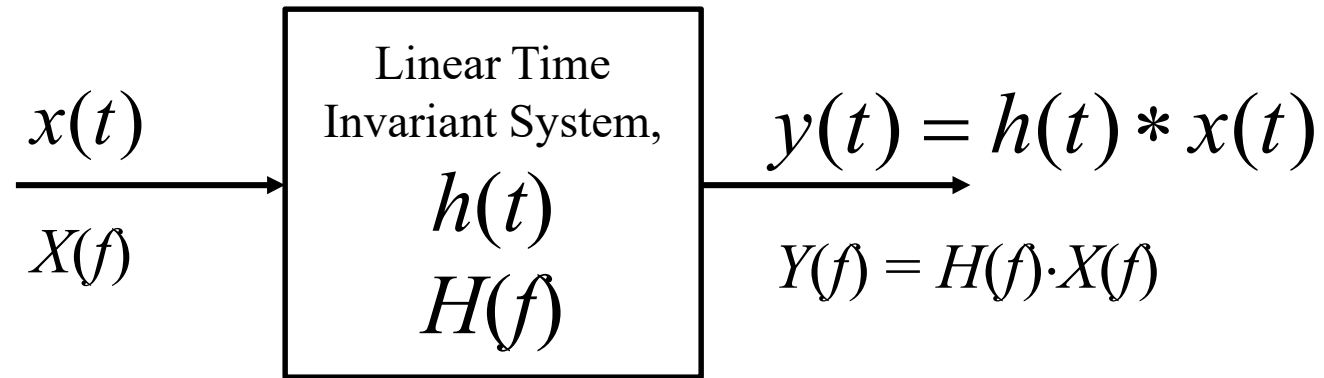
Data Communication

Instructor:

Dr. Md. Monirul Islam

Analysis of Signals and Systems

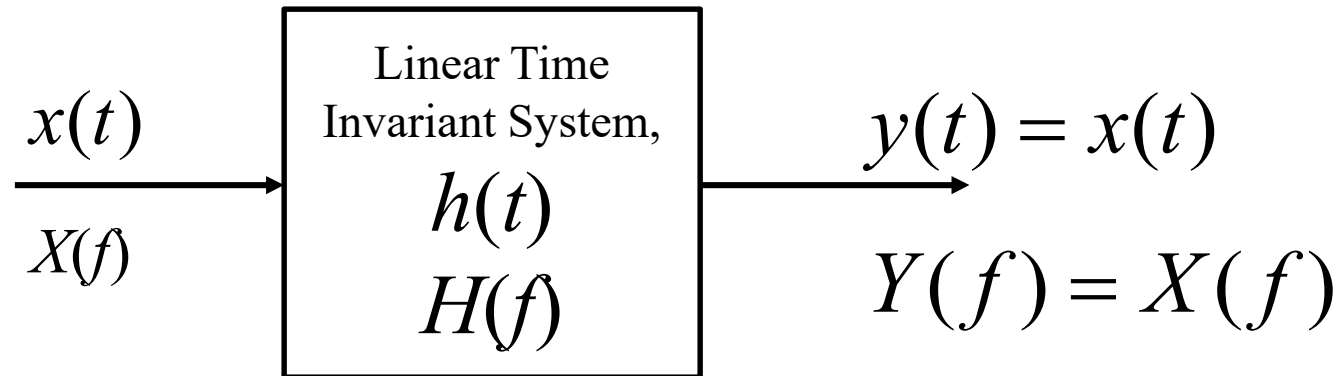
Signal Distortion during Transmission



$$|Y(f)|e^{j\theta_y(f)} = |H(f)||X(f)|e^{j[\theta_h(f)+\theta_x(f)]}$$

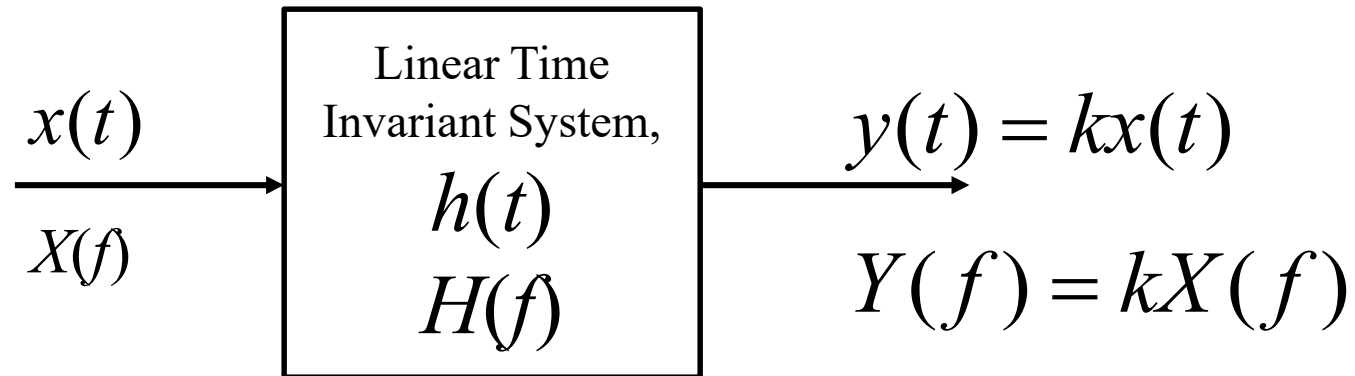
- System distorts the input signal
- Input signal amplitude $|X(f)|$ is changed to $|X(f)| |H(f)|$
- Input signal phase is shifted by $\theta_h(f)$
- Plot of $|H(f)|$ and $\theta_h(f)$ vs. f shows how the system changes different frequency components differently (attenuation/boosting)

Distortionless Transmission



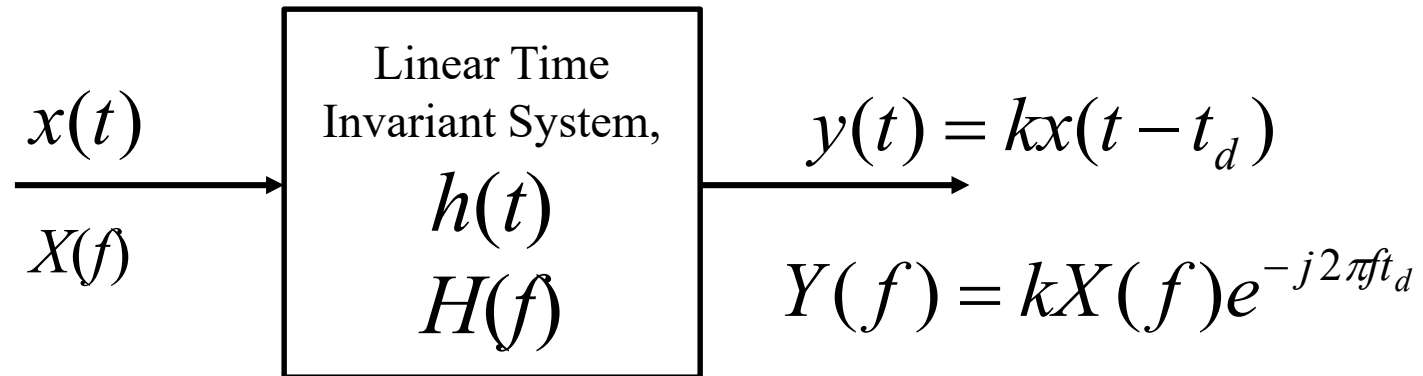
- Input and output signals have identical shape

Distortionless Transmission



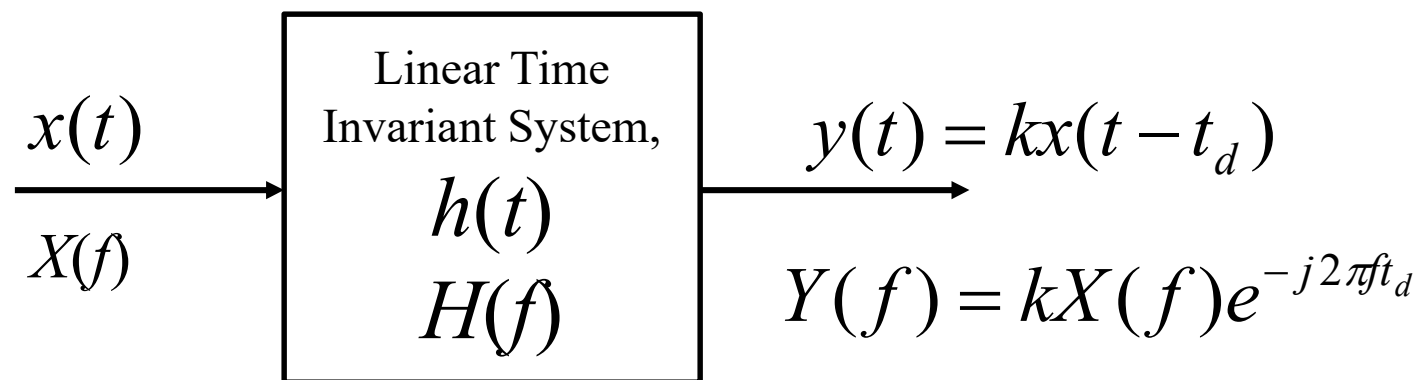
- Input and output signals have identical shape
- Boosting/amplification is allowed

Distortionless Transmission



- Input and output signals have identical shape
- Boosting/amplification is allowed
- Delaying is allowed

Distortionless Transmission

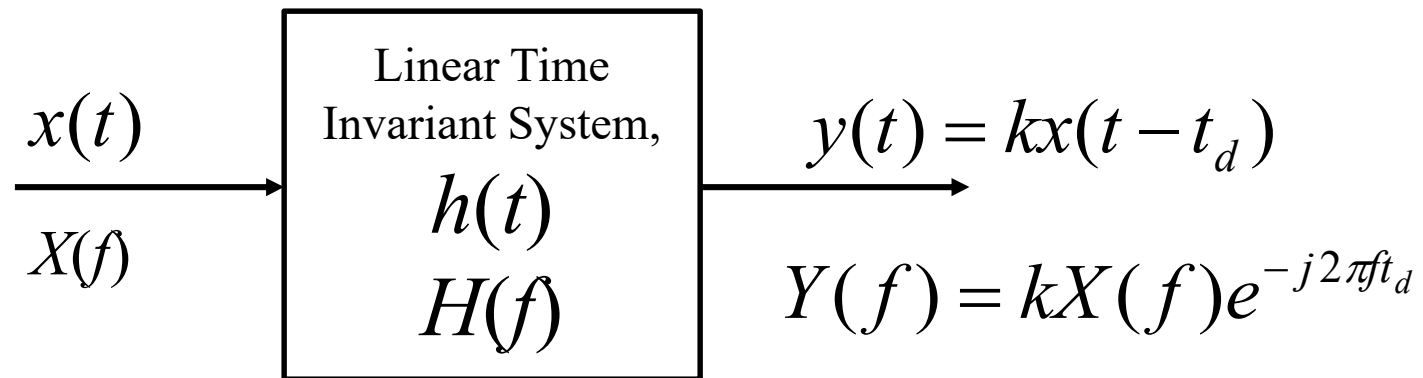


- Input and output signals have identical shape
- Delaying is allowed

$$Y(f) = kX(f)e^{-j2\pi ft_d} = ke^{-j2\pi ft_d} X(f) = H(f)X(f)$$

$$H(f) = ke^{-j2\pi ft_d}$$

Distortionless Transmission

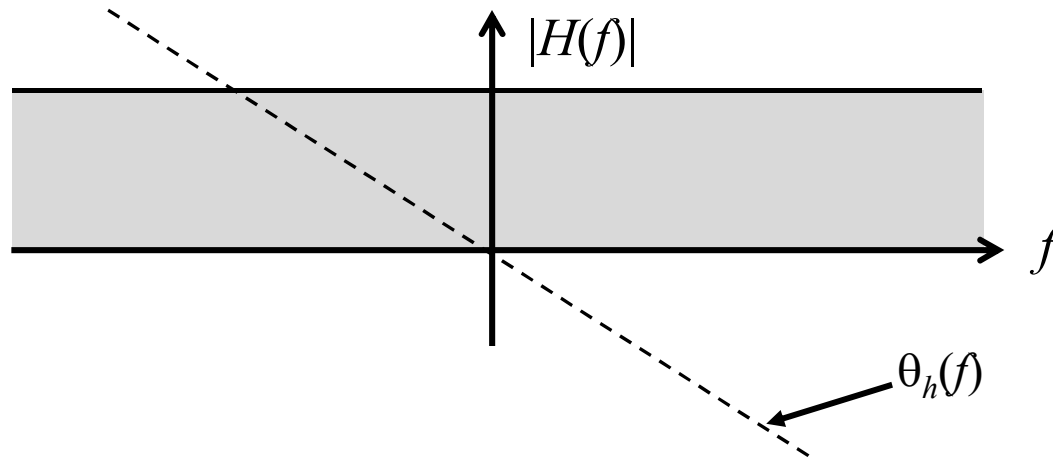


$$H(f) = ke^{-j2\pi ft_d}$$

$$|H(f)| = k \quad \text{and} \quad \theta_h(f) = -2\pi ft_d$$

- $|H(f)|$ is constant
- $\theta_h(f)$ is a **linear function** of f

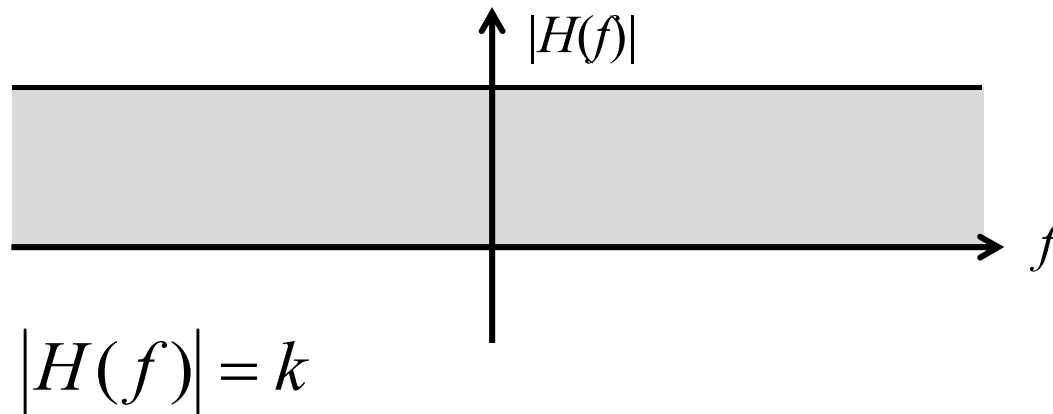
Distortionless Transmission



$$|H(f)| = k \quad \text{and} \quad \theta_h(f) = -2\pi f t_d = -\omega t_d$$

- $|H(f)|$ is constant
- $\theta_h(f)$ is a **linear function** of f
- Slope of $\theta_h(\omega)$ is $-t_d$, where t_d is delay of the output

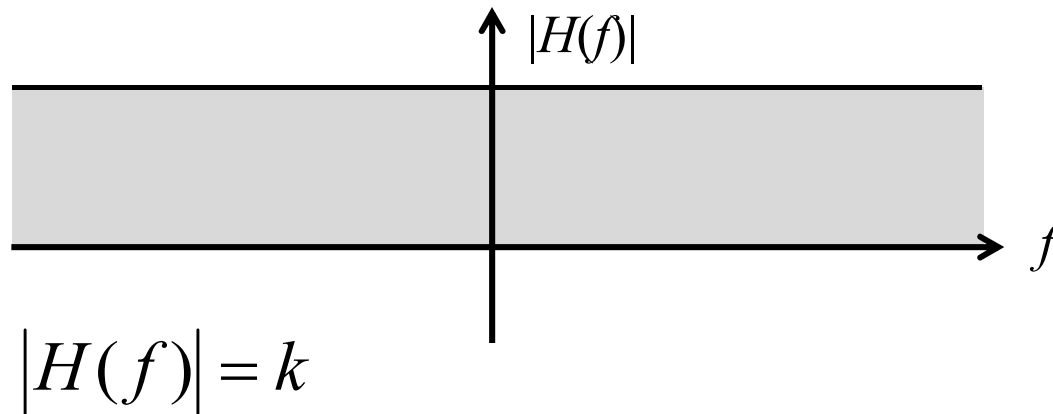
All Pass Vs. Distortionless System



$$|Y(f)| = |H(f)| |X(f)|$$

- $|H(f)| = k$ means that every frequency amplitude of $|X(f)|$ is passed to $|Y(f)|$ **with same gain**
- Everything is present $|Y(f)|$, nothing is missing!

All Pass Vs. Distortionless System



$$|Y(f)| = |H(f)| |X(f)|$$

- $|H(f)| = k$ means that every frequency amplitude of $|X(f)|$ is passed to $|Y(f)|$ **with same gain**
- Everything is present $|Y(f)|$, nothing is missing!
- **However, this does NOT guarantee distortionless system**

All Pass Vs. Distortionless System

All pass system:

$$|H(f)| = k$$

Consider a **real composite** signal consisting of **multiple sinusoids** of **different frequencies**

- $|H(f)| = k$ guarantees only presence of all frequency components
- **Different sinusoid** may get **different delays**.
- Synchronization of multiple frequency components is NOT guaranteed
- Output composite signal will be **out of sync**

All Pass Vs. Distortionless System

Distortionless system:

$$|H(f)| = k \quad \text{and} \quad \theta_h(f) = -2\pi f t_d$$

- Linear phase $\theta_h(f) = -2\pi f t_d$ guarantees a constant time delay t_d for every sinusoid
- Synchronization is guaranteed

All Pass Vs. Distortionless System

How to Check Distortionless transmission?

Find the slope (derivative) of $\theta_h(f)$

$$t_d(f) = -\frac{1}{2\pi} \cdot \frac{d\theta_h(f)}{df}$$

All Pass Vs. Distortionless System

How to Check Distortionless transmission?

Find the slope (derivative) of $\theta_h(f)$

$$t_d(f) = -\frac{1}{2\pi} \cdot \frac{d\theta_h(f)}{df}$$

Hint: Rearrange and differentiate $\theta_h(f) = -2\pi f t_d$

All Pass Vs. Distortionless System

How to Check Distortionless transmission?

Find the slope (derivative) of $\theta_h(f)$

$$t_d(f) = -\frac{1}{2\pi} \cdot \frac{d\theta_h(f)}{df}$$

- If the slope of $\theta_h(f)$ is constant, it guarantees a constant time delay t_d for every sinusoid
- If the slope is a function of f , different frequency components will have different time delay

Nature of Distortion in Audio Signals

Human ear is

- **sensitive** to amplitude distortion which results if $|H(f)| \neq k$
- **insensitive** to phase distortion which results if the slope of $\theta_h(f)$ is NOT constant

Nature of Distortion in Audio Signals

- Duration of each spoken syllable : 0.01 to 0.1 second
- Variation of the slope of $\theta_h(f)$ is ONLY fraction of a millisecond, meaning a nearly constant slope, e.g., time delay

Nature of Distortion in Video Signals

Human eye is

- insensitive to amplitude distortion
- sensitive to phase distortion

Nature of Distortion in Video Signals

Insensitive amplitude distortion

- amplitude distortion changes only half-tone values of picture

Sensitive phase distortion

- Nonlinear phase distortion means different time delay for different picture (signal) element
- This results in smeared picture

Nature of Distortion in Communication Systems

Sensitive phase distortion

- Nonlinear phase distortion causes **pulse dispersion** (spreading out)
- **Pulses are interfered** with near by pulses
- This results in misinterpretation in the receiver
 - Binary **1** may read as **0** and vice versa

Ideal Vs. Practical Filters

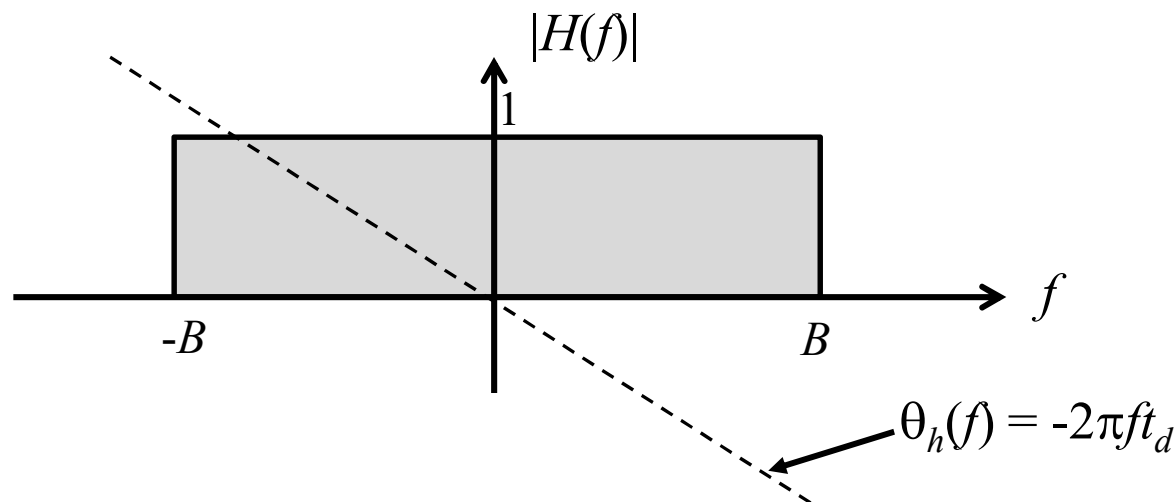
Ideal filters

- allow distortionless transmission for certain band of frequencies
- suppress all other frequencies

Ideal Vs. Practical Filters

This Ideal Lowpass filter

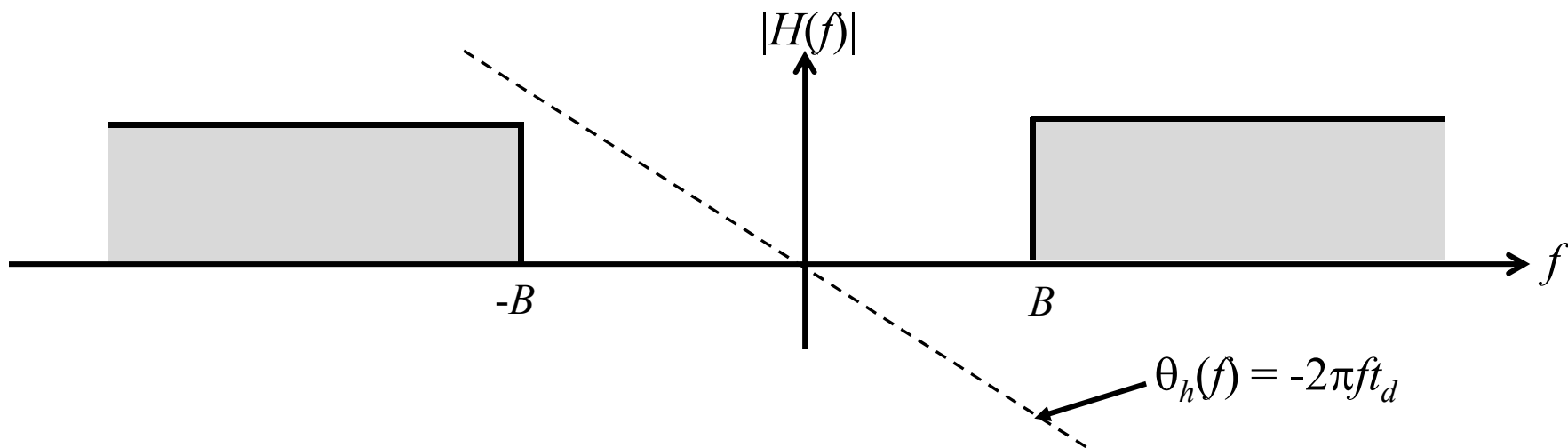
- allows transmission for frequencies $f \leq |B|$
- suppresses all other frequencies $f \geq |B|$



Ideal Vs. Practical Filters

This Ideal Highpass filter

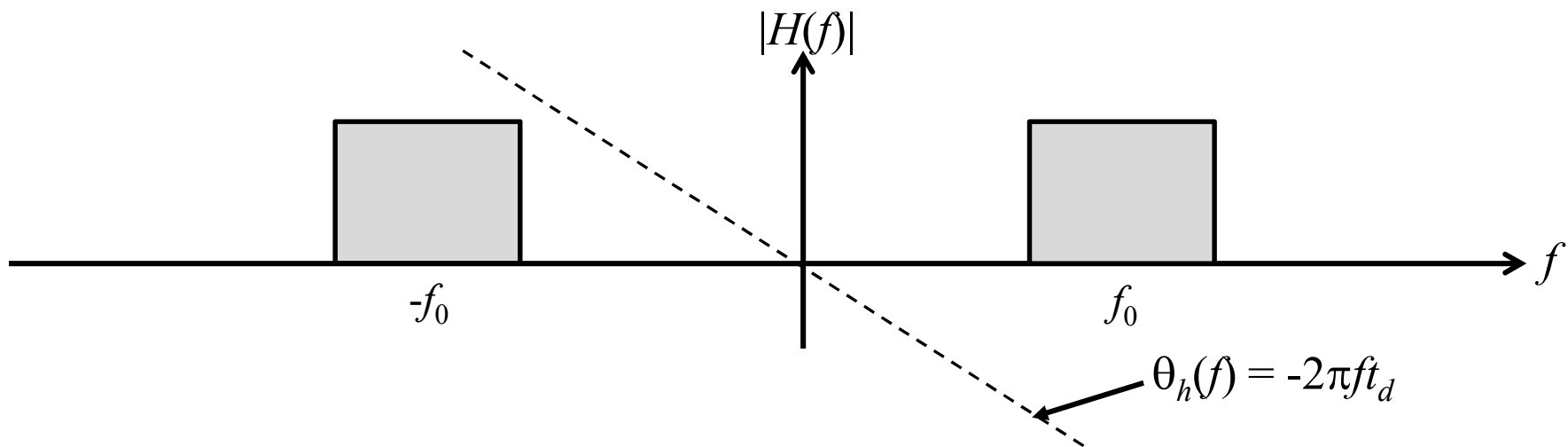
- allows transmission for frequencies $f \geq |B|$
- suppresses all other frequencies $f \leq |B|$



Ideal Vs. Practical Filters

This Ideal Bandpass filter

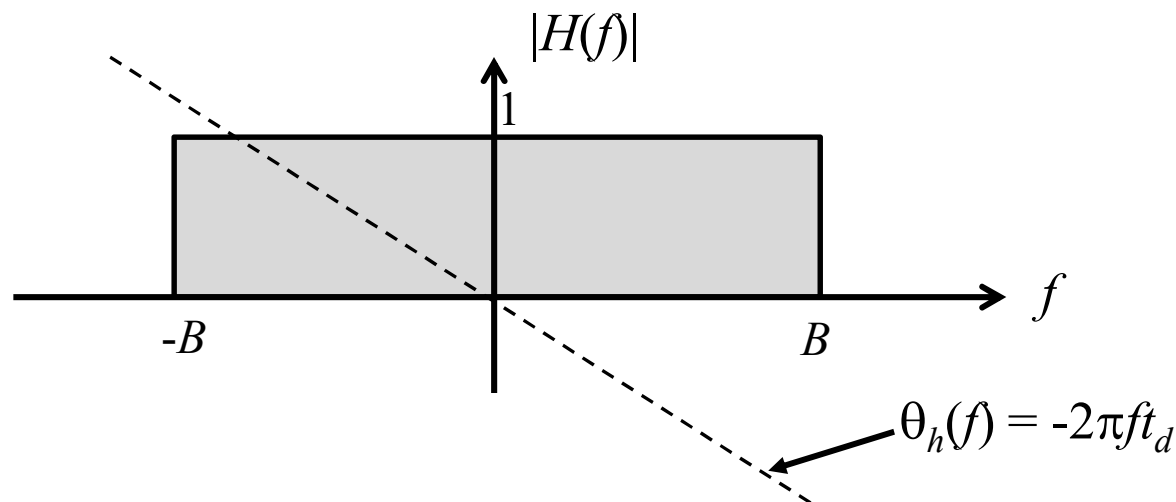
- allows transmission for the band centered at $\pm f_0$
- suppresses all other frequencies



Ideal Vs. Practical Filters

This Ideal Lowpass filter

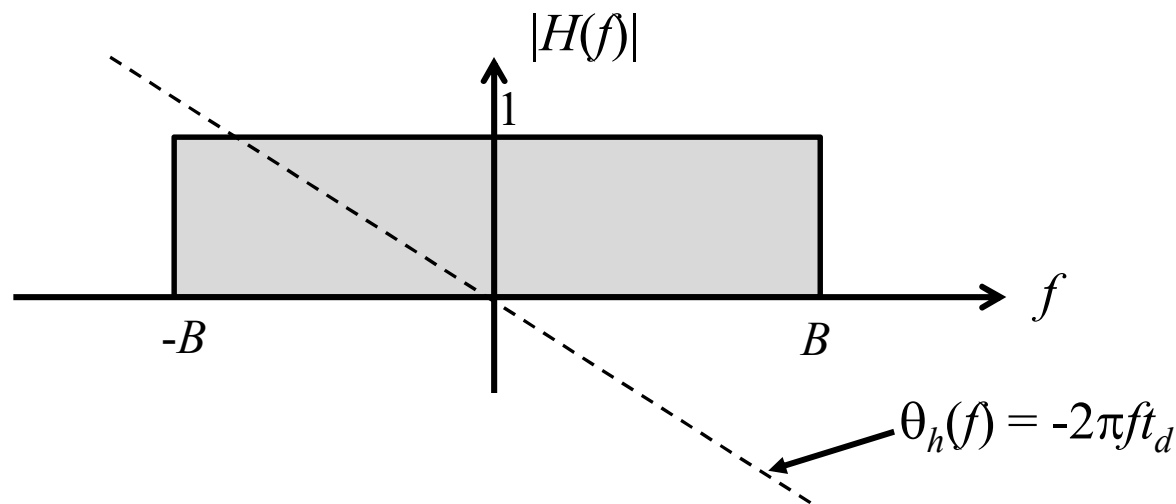
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Ideal Vs. Practical Filters

This Ideal Lowpass filter

- allows transmission for frequencies $f \leq |B|$
- suppresses all other frequencies $f \geq |B|$
- Slope of $\theta_h(f)$ is constant $-t_d$, means time delay is t_d

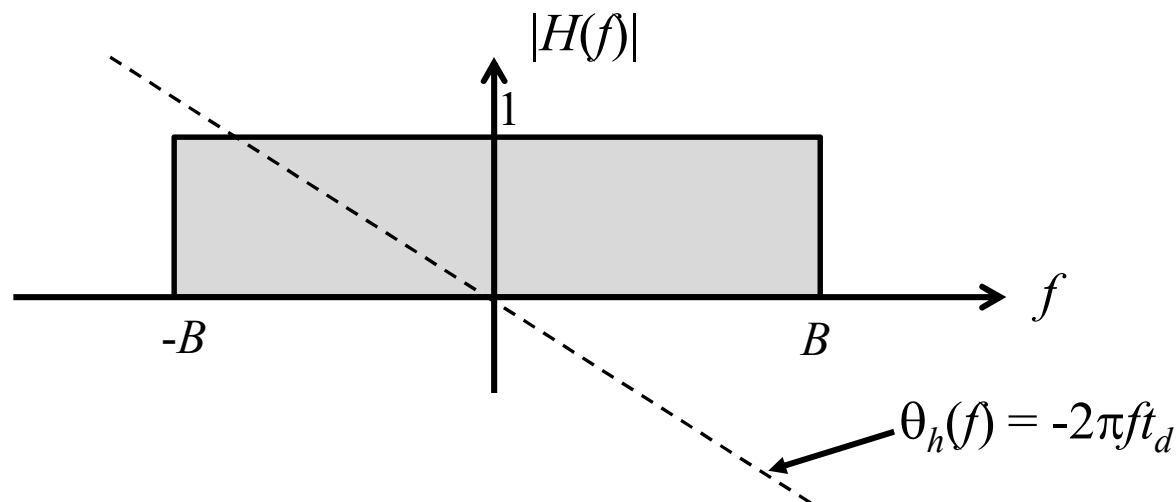


Ideal Vs. Practical Filters

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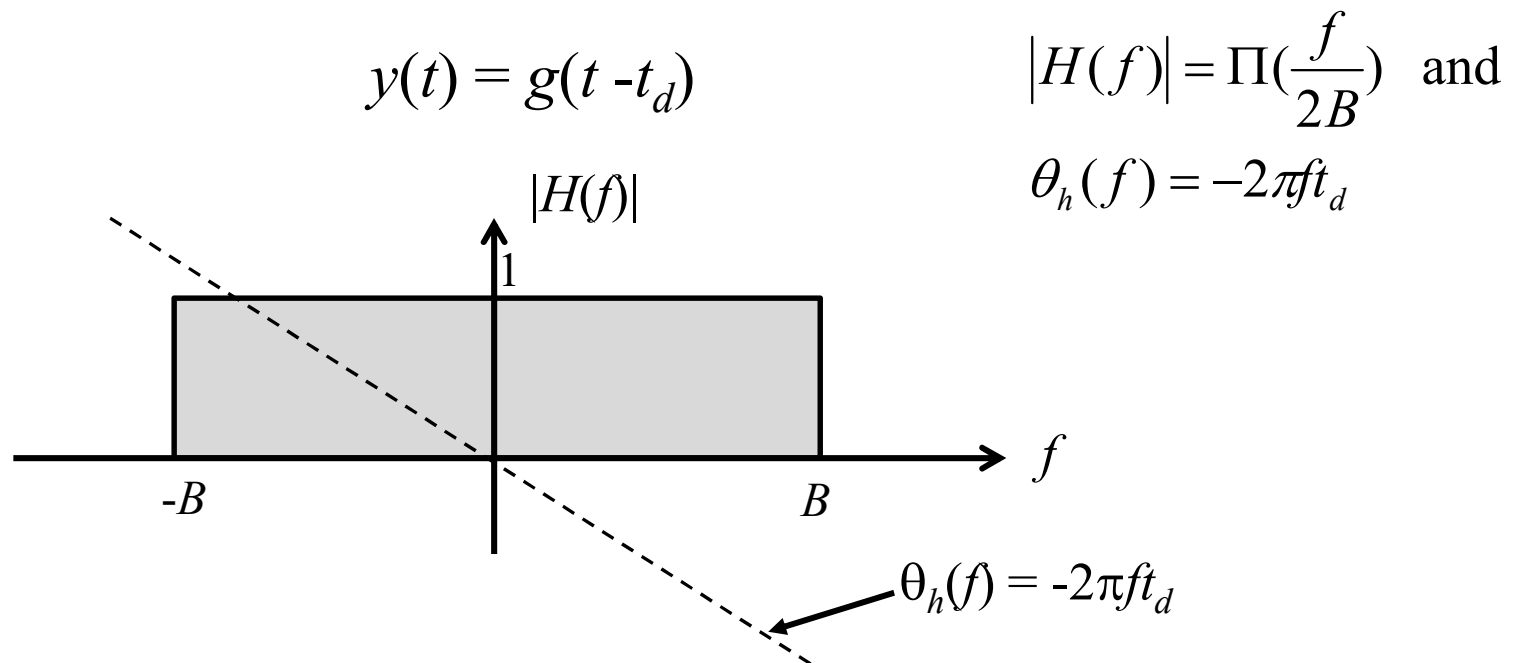
$$y(t) = g(t - t_d)$$



Ideal Vs. Practical Filters

This Ideal Lowpass filter

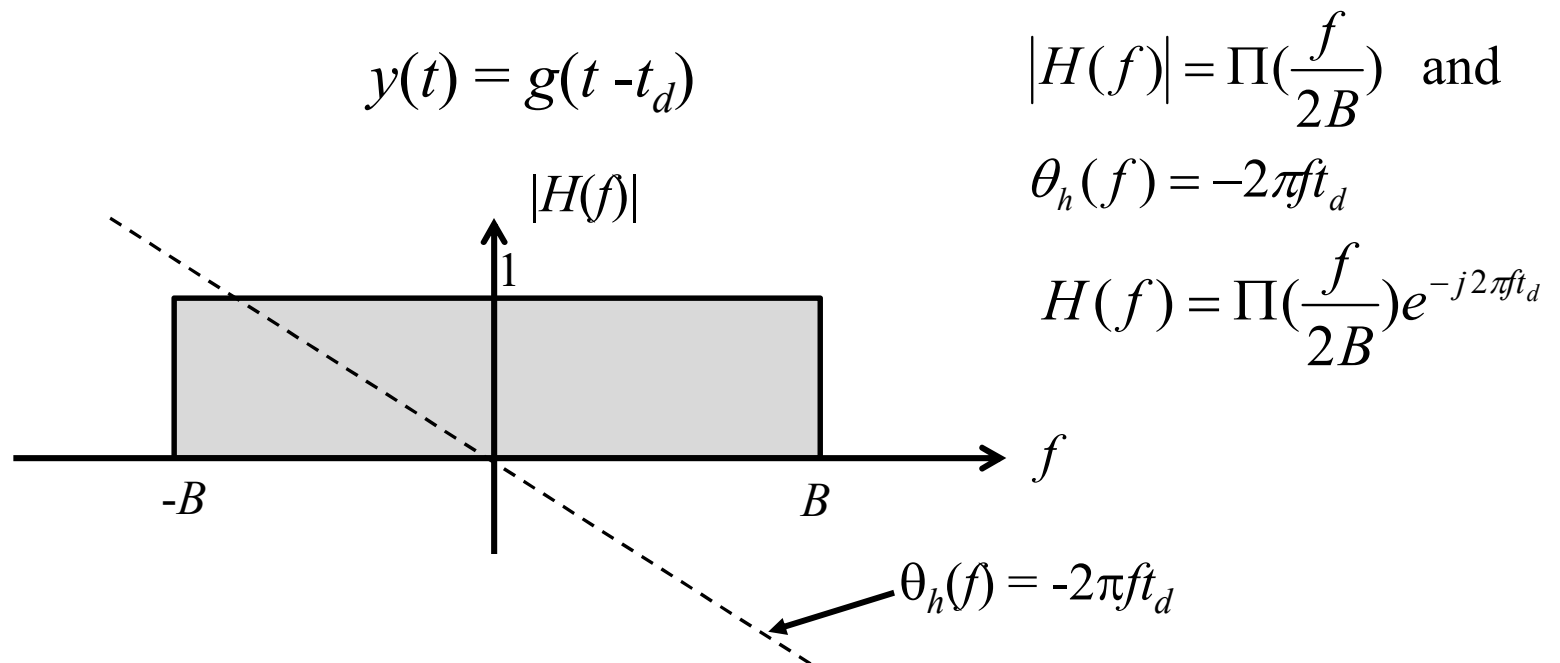
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Ideal Vs. Practical Filters

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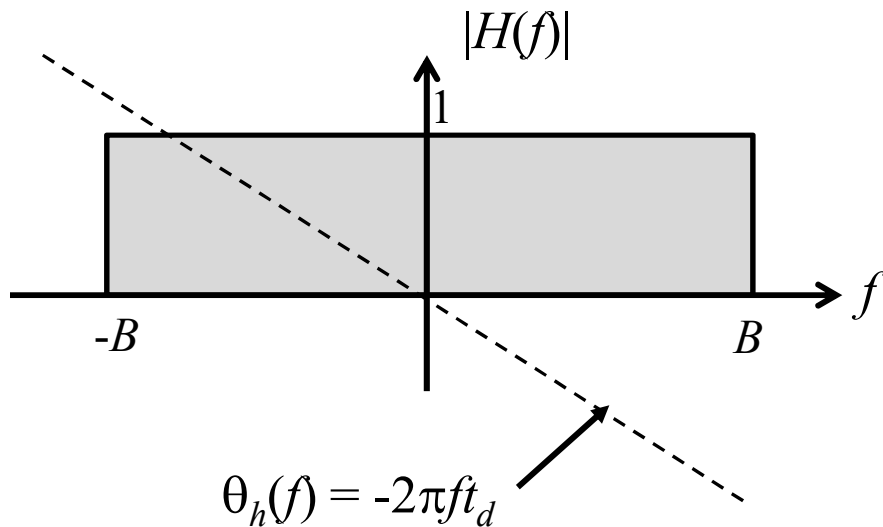


Ideal Vs. Practical Filters

Impulse response of Ideal Lowpass filter

Frequency Transform of
Impulse response

$$H(f) = \Pi\left(\frac{f}{2B}\right)e^{-j2\pi ft_d}$$



Ideal Vs. Practical Filters

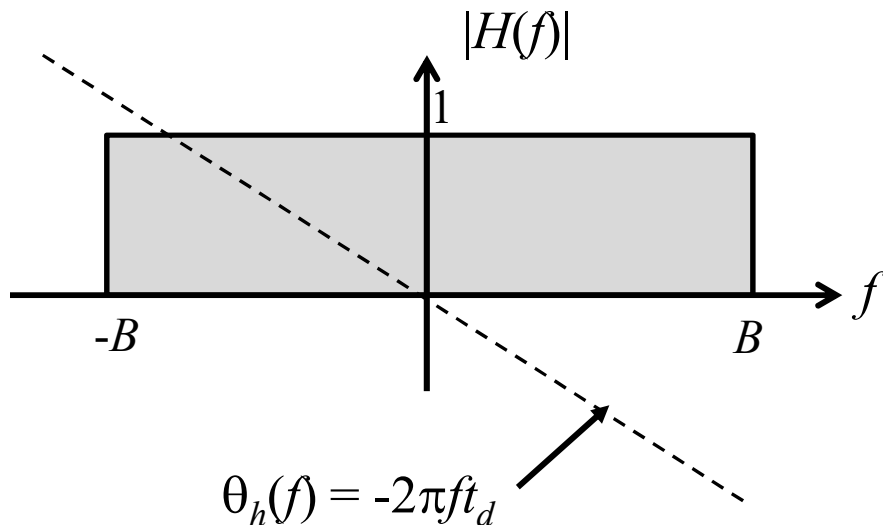
Impulse response of Ideal Lowpass filter

Frequency Transform of
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$$H(f) = \Pi\left(\frac{f}{2B}\right)e^{-j2\pi ft_d}$$

Impulse response

$$\begin{aligned} h(t) &= \mathfrak{F}^{-1}[H(f)] = \mathfrak{F}^{-1}\left[\Pi\left(\frac{f}{2B}\right)e^{-j2\pi ft_d}\right] \\ &= 2B\text{sinc}(2\pi B(t - t_d)) \end{aligned}$$

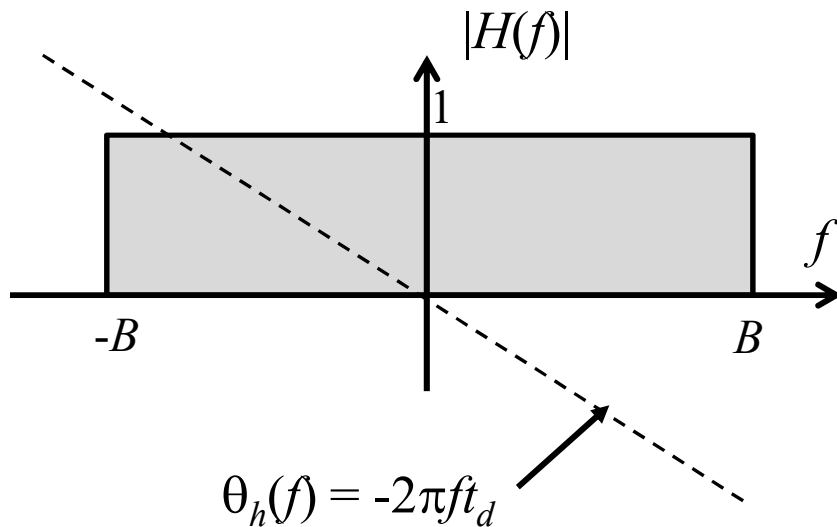


Ideal Vs. Practical Filters

Impulse response of Ideal Lowpass filter

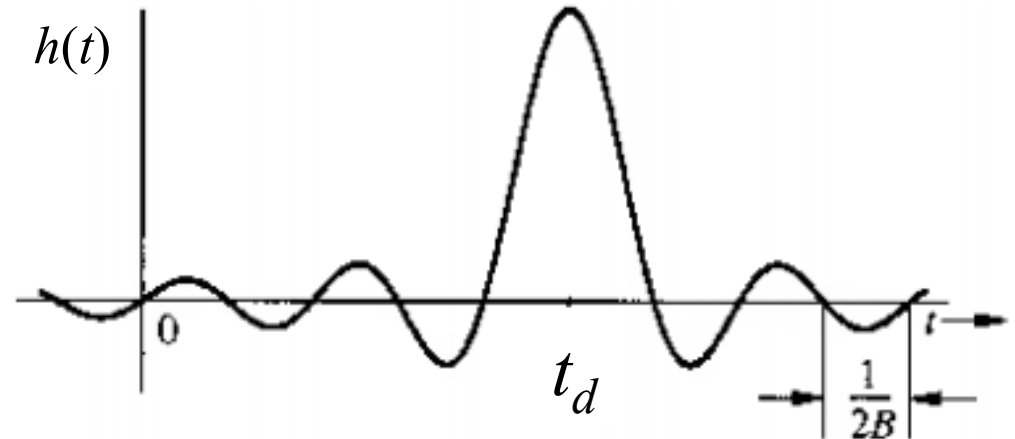
Frequency Transform of
Impulse response

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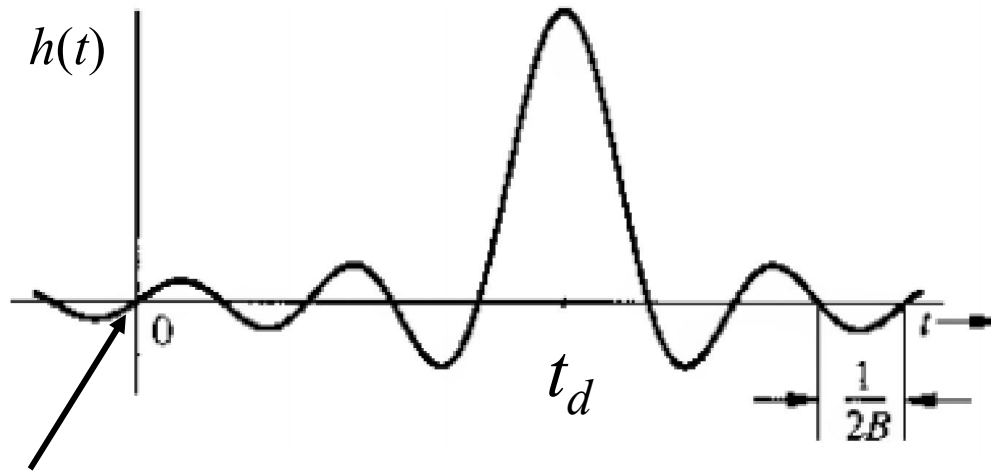
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Ideal Vs. Practical Filters

Impulse response of Ideal Lowpass filter

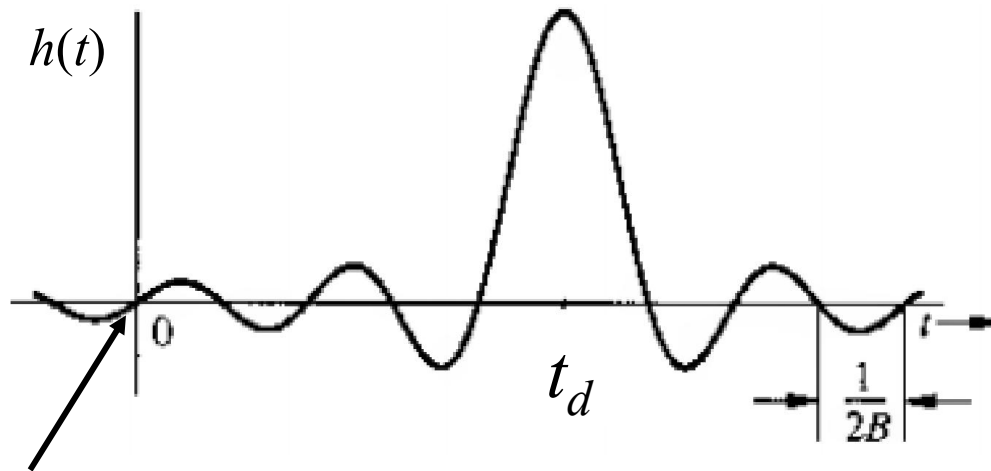


Impulse response
 $h(t) = 2B \text{sinc}(2\pi B(t - t_d))$

Response begins
before $t = 0$

Ideal Vs. Practical Filters

Impulse response of Ideal Lowpass filter



Impulse response

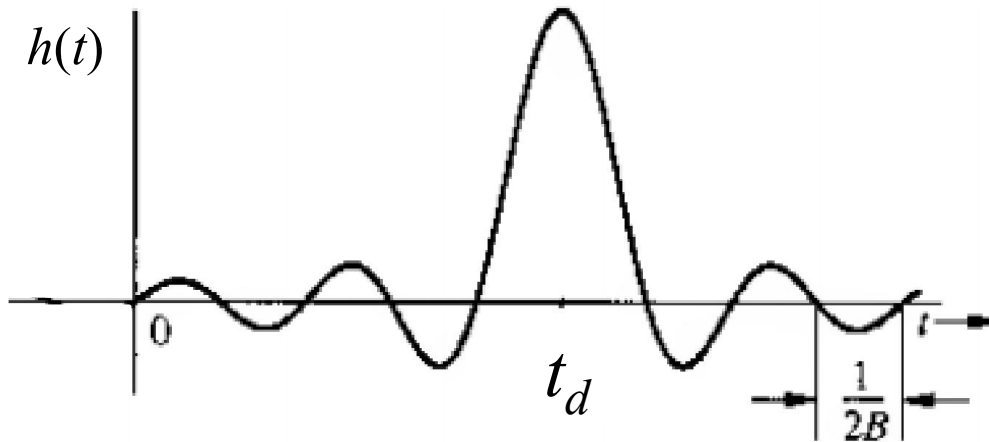
$$h(t) = 2B \text{sinc}(2\pi B(t - t_d))$$

Response begins
before $t = 0$

Noncausal and
unrealizable filter

Ideal Vs. Practical Filters

Causal and realizable Lowpass filter



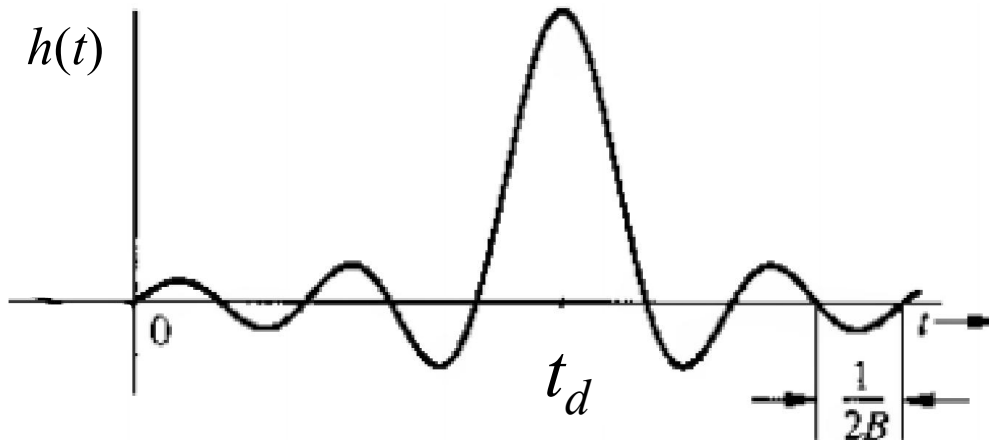
Condition in time domain

$$h(t) = 0 \text{ for all } t < 0$$

Response begins
on or after $t = 0$

Ideal Vs. Practical Filters

Causal and realizable Lowpass filter



Response begins
on or after $t = 0$

Condition in time domain

$$h(t) = 0 \text{ for all } t < 0$$

Condition in frequency
domain

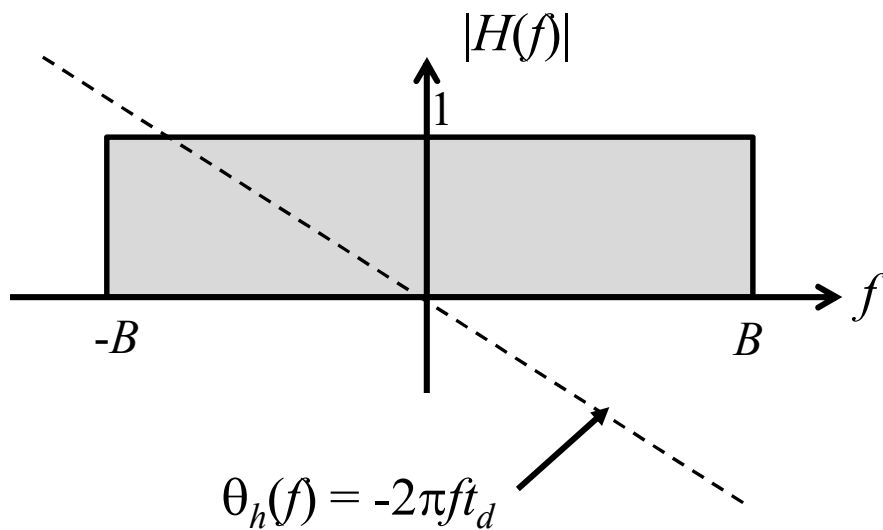
$$\int_{-\infty}^{\infty} \left(\frac{|\alpha(f)|}{1 + f^2} \right) df < \infty$$

$$\text{where } \alpha(f) = \ln|H(f)|$$

Paley-Wiener Criterion

Ideal Vs. Practical Filters

Causal and realizable Lowpass filter



Condition in frequency domain

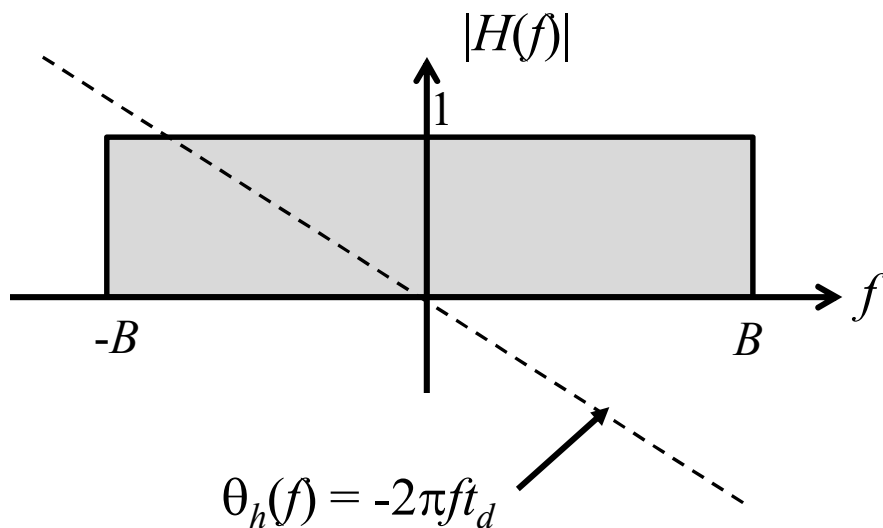
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Paley-Wiener Criterion

Ideal Vs. Practical Filters

Causal and realizable Lowpass filter



Condition in frequency domain

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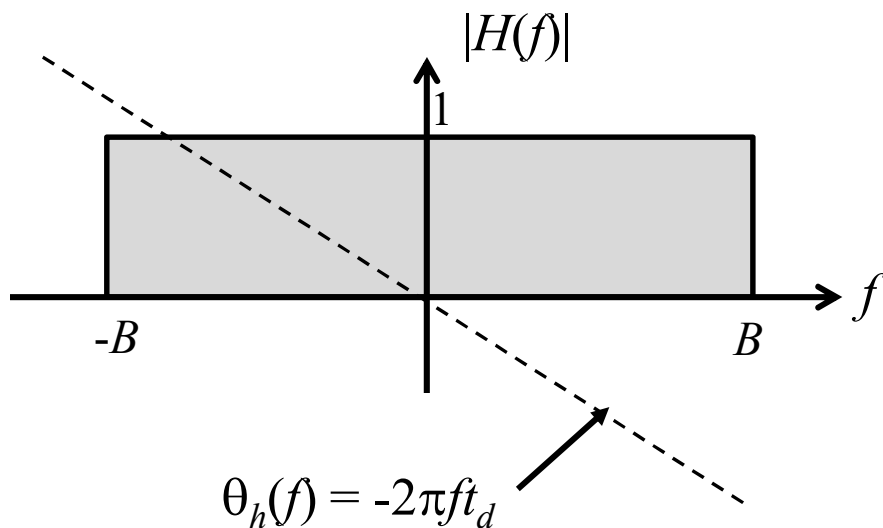
where $\alpha(f) = \ln |H(f)|$

Paley-Wiener Criterion

- If for **certain band** of frequencies $|H(f)| = 0$ means $|\alpha(f)| = \ln |H(f)| = \infty$, which does **NOT** satisfy P-W criterion

Ideal Vs. Practical Filters

Causal and realizable Lowpass filter



Condition in frequency domain

$$\int_{-\infty}^{\infty} \left(\frac{|\alpha(f)|}{1 + f^2} \right) df < \infty$$

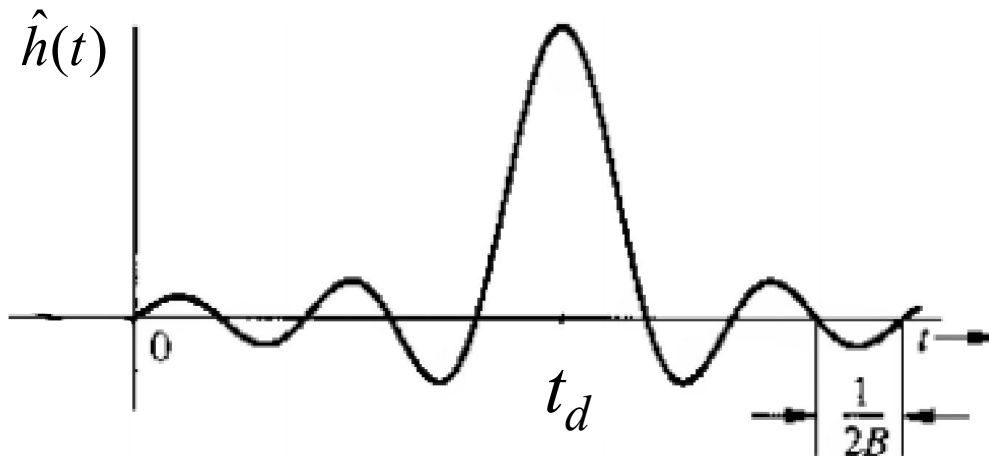
where $\alpha(f) = \ln|H(f)|$

Paley-Wiener Criterion

- If for **discrete** of frequencies $|H(f)| = 0$, P-W criterion may be satisfied

Ideal Vs. Practical Filters

Causal and realizable Lowpass filter



$$\hat{h}(t) = h(t)u(t)$$

Condition in frequency domain

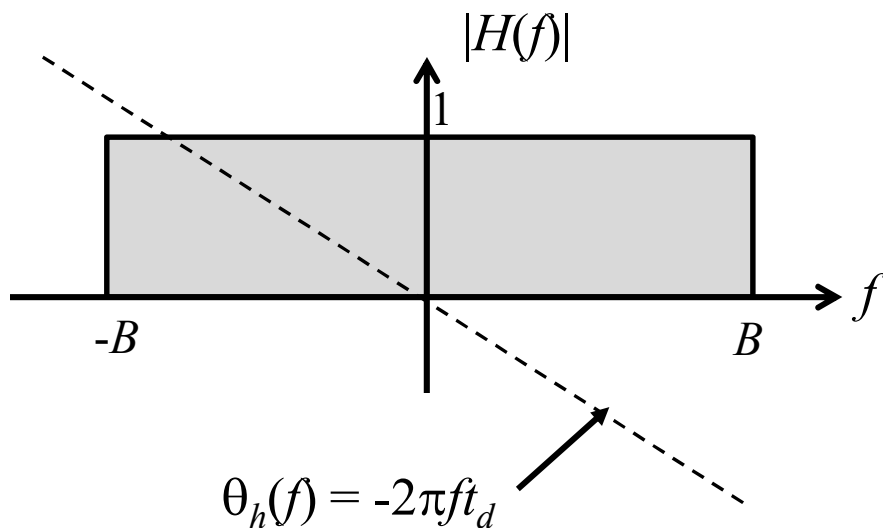
$$\int_{-\infty}^{\infty} \left(\frac{|\alpha(f)|}{1 + f^2} \right) df < \infty$$

where $\alpha(f) = \ln|H(f)|$

Paley-Wiener Criterion

Ideal Vs. Practical Filters

Other Causal and realizable Lowpass filters



Condition in frequency domain

$$\int_{-\infty}^{\infty} \left(\frac{|\alpha(f)|}{1 + f^2} \right) df < \infty$$

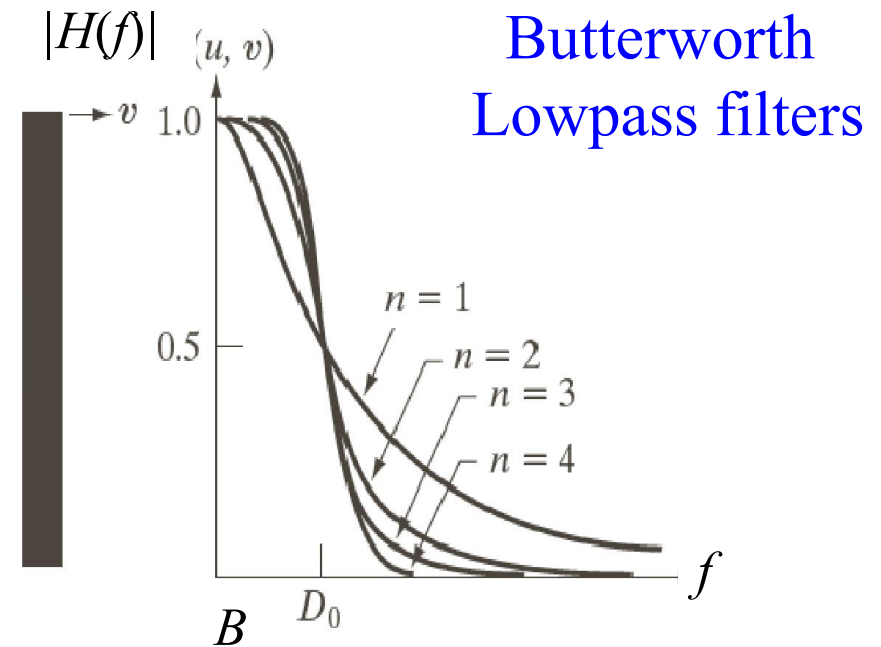
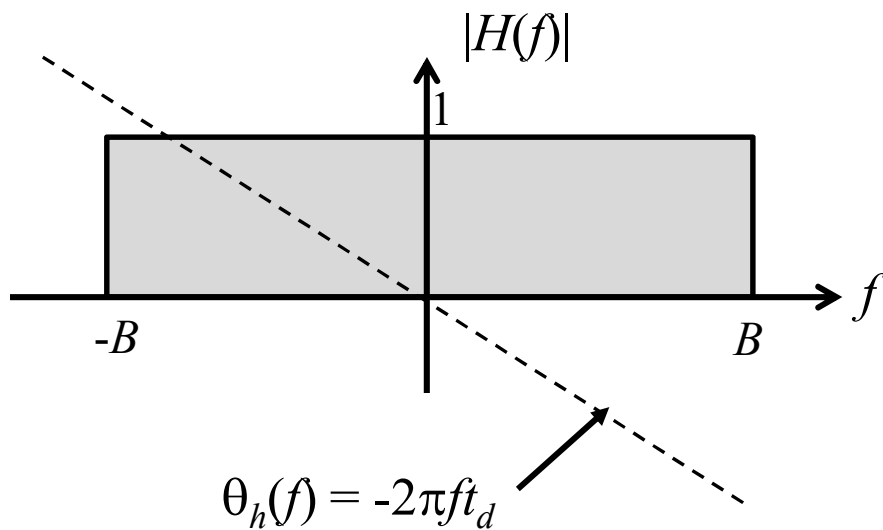
where $\alpha(f) = \ln|H(f)|$

Paley-Wiener Criterion

- Instead of sharp change at $\neq B$, change smoothly and approach asymptotically to zero

Ideal Vs. Practical Filters

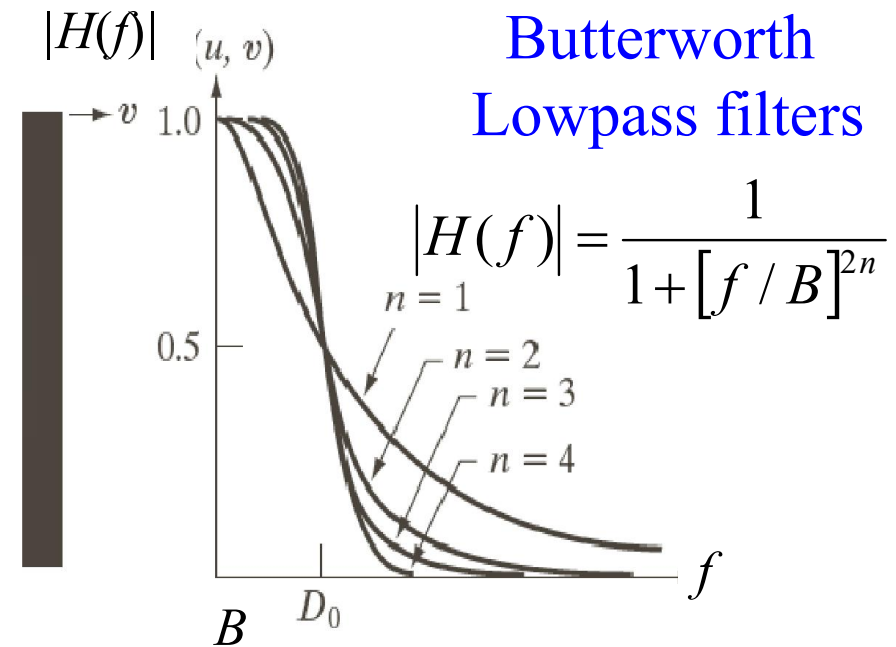
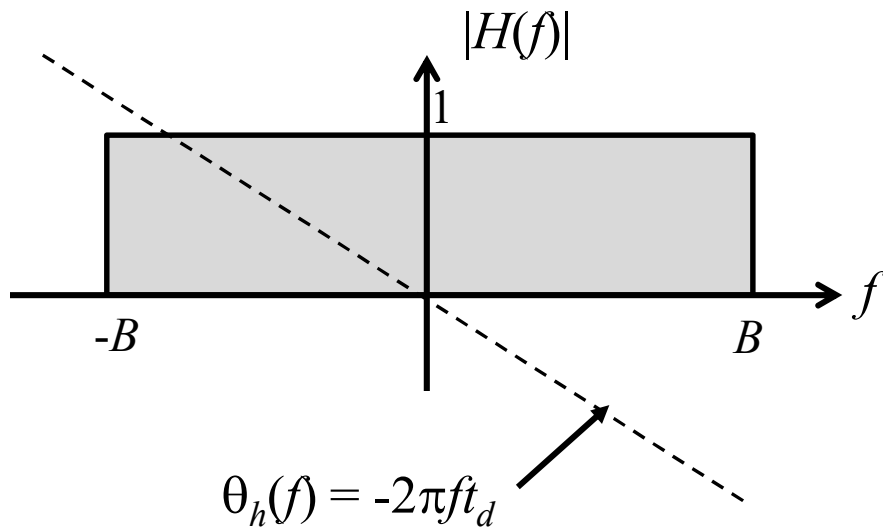
Other Causal and realizable Lowpass filters



- Instead of sharp change at $\neq B$, change smoothly and approach asymptotically to zero

Ideal Vs. Practical Filters

Other Causal and realizable Lowpass filters



- Instead of sharp change at $\pm B$, change smoothly and approach asymptotically to zero